

Sampling:

$$\begin{aligned}x_p(t) &= x(t) \cdot p(t) \\&= x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT)\end{aligned}$$

$$x_p(t) = \sum_n x(nt) \delta(t - nT)$$

We want $X_p(j\omega) \rightarrow$ I forgot why.

Read 1: $X_p(j\omega) = F(x_p(t)) = \int_{-\infty}^{\infty} \sum_n x(nt) \delta(t - nT) e^{-j\omega t} dt$

$$X_p(j\omega) = \sum_{n=-\infty}^{\infty} x(nt) e^{-j\omega nT}$$

Read 2:

$$X_p(j\omega) = \frac{1}{2\pi} [X(j\omega) * P(j\omega)] \quad \left| \quad P(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s) \right.$$

$$\therefore X_p(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s))$$

Nyquist - Shannon Theorem:

Each copy of $X(j\omega)$ at $k\omega_s$ occupies $(k\omega_s - \omega_m, k\omega_s + \omega_m)$

So, to avoid overlap,

we need $\omega_s > 2\omega_m$

Oversampling: $\omega_s > 2\omega_m$

Undersampling: $\omega_s < 2\omega_m$

or Aliasing

$$x_r(t) = \sum_{n=-\infty}^{\infty} x[n] \operatorname{sinc}\left(\frac{t-nT}{T}\right)$$

$$\operatorname{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$$